PHASE 3 — TEST CASE DOCUMENTATION

AI vs AI — Autonomous Cyber Defense Challenge

Team	Code Titans (UG - 55)
Domain	Cybersecurity
Track	Track 01
Members	Abhishek, Daiwik, Jeevan, Prajwal
Project	ATLAS Security Operations Dashboard

Purpose: Document repeatable tests for major project features, validation, security controls, and usability.

Technology: React + TypeScript, Python FastAPI, SQLite; repository includes pytest and Playwright tests.

Execution status: These cases have not been executed while preparing this PDF. Actual results are marked not executed and status is Not Run. Update these fields after testing; do not claim a pass without evidence.

Safety scope: Use synthetic/replay data and sandbox state. No real attacks, remote command execution, or real containment actions are intended.

TEST CASES

Test Case ID	Feature / Functionality	Test Input	Expected Result	Actual Result	Pass/Fail	Justification
TC-01	Dashboard initialization	Open app with clean test database.	Dashboard loads; overview metrics and synthetic replay events are visible.	Not executed during preparation.	Not Run	Confirms app starts in a usable state.
TC-02	Manual patrol	Click Run Patrol with replay events loaded.	Patrol completes or reports a clear error; history/coverage updates; recommendations remain pending.	Not executed during preparation.	Not Run	Validates detection without automatic response execution.
TC-03	Incident investigation	Open an incident and select Investigate.	A summary, evidence, limitations, and related event IDs are returned.	Not executed during preparation.	Not Run	Makes investigation results traceable to evidence.
TC-04	Human approval gate	Investigate an incident; do not approve recommendations.	Recommendations stay Pending; sandbox state does not change before approval.	Not executed during preparation.	Not Run	Preserves human oversight of response actions.
TC-05	Reject recommendation	Reject an action with analyst name and reason.	Action is marked rejected and is not executed.	Not executed during preparation.	Not Run	Confirms analysts can decline proposed responses.
TC-06	Prevent decision override	Reject an action, then try to approve it.	Second decision is rejected (expected HTTP 409); rejection remains in force.	Not executed during preparation.	Not Run	Prevents contradictory decisions from changing action state.
TC-07	Approval input validation	Attempt approval without confirmation or required analyst details.	Request is rejected; action remains pending; no state change occurs.	Not executed during preparation.	Not Run	Requires confirmation and analyst accountability.
TC-08	Create simulated server	Add Server > Simulated Demo Source; enter valid name and unique hostname.	Server is created, enrollment instructions appear, telemetry is labelled simulated.	Not executed during preparation.	Not Run	Verifies demo enrollment and data provenance.
TC-09	Required server fields	Create a server with name or hostname empty.	Validation rejects request; no server record is created.	Not executed during preparation.	Not Run	Avoids incomplete registrations.
TC-10	Revoke server credentials	Revoke a test server; submit telemetry with old agent token.	Server status becomes revoked; old token is rejected (expected HTTP 401).	Not executed during preparation.	Not Run	Stops revoked collectors from sending telemetry.
TC-11	Valid website connector	Enter valid HTTPS fixture endpoint and API key; test connection.	Connection verifies only when authentication and supported API contract pass.	Not executed during preparation.	Not Run	A URL alone must not prove connectivity.
TC-12	Invalid website credentials	Enter invalid API key and test connection.	Authentication fails clearly; saving stays disabled until verification succeeds.	Not executed during preparation.	Not Run	Prevents saving an unverified connector.
TC-13	Website permission validation	Verify connector, change a permission/category, then save.	Required checks are repeated; saved permissions match verified grants.	Not executed during preparation.	Not Run	Ensures access matches granted permissions.
TC-14	Telemetry filters	Filter events by source, severity/campaign, and matching text.	Only matching events appear; clearing filters restores broader list.	Not executed during preparation.	Not Run	Helps analysts isolate relevant telemetry.
TC-15	Pause event display	Pause visible feed while patrols continue, then resume.	Visible snapshot stays stable while paused; resume refreshes it; backend patrols continue.	Not executed during preparation.	Not Run	Separates pausing display from stopping processing.
TC-16	JSON export	Apply filter and export telemetry as JSON.	Downloaded JSON parses and contains only filtered records.	Not executed during preparation.	Not Run	Supports portable investigation and analysis.

Test Case ID	Feature / Functionality	Test Input	Expected Result	Actual Result	Pass/Fail	Justification
TC-17	CSV export safety	Export records with commas, quotes, newlines, and formula-leading text.	CSV is escaped correctly and formula-leading cells are neutralized.	Not executed during preparation.	Not Run	Reduces malformed export and spreadsheet formula risks.
TC-18	Persistence after restart	Create source/server; restart backend with same SQLite database.	Persisted records remain available and consistent.	Not executed during preparation.	Not Run	Checks normal restart persistence.
TC-19	Missing session token	Call protected API endpoint without required session token.	Request is denied (expected HTTP 401/403 per endpoint contract); no protected data exposed or changed.	Not executed during preparation.	Not Run	Verifies backend authentication enforcement.
TC-20	Schedule minimum boundary	Attempt patrol interval below 30 seconds.	Invalid interval is rejected; valid settings remain unchanged.	Not executed during preparation.	Not Run	Prevents unsupported scheduling intervals.
TC-21	Schedule maximum boundary	Set interval to 24 hours, then try above 24 hours.	24 hours is accepted; values above it are rejected.	Not executed during preparation.	Not Run	Checks upper boundary and out-of-range rejection.
TC-22	Mobile layout	Open dashboard at 390 × 844 viewport and inspect controls.	Primary controls remain usable; unintended horizontal overflow is absent.	Not executed during preparation.	Not Run	Checks mobile-sized usability.
TC-23	Modal keyboard dismissal	Open Connect Data Source modal and press Escape.	Modal closes where supported; page remains usable and no browser error occurs.	Not executed during preparation.	Not Run	Checks standard keyboard dialog dismissal.
TC-24	Source data isolation	Register two test sources and inspect events/incidents.	Each source's data and credentials remain associated with the correct source.	Not executed during preparation.	Not Run	Prevents cross-source attribution errors and data leakage.

EXECUTION SUMMARY

Total cases	Passed	Failed	Not Run
24	0 — not executed	0 — not executed	24

End of document.